Statistics PAST Paper

health records and information technology (Kenya Medical Training College)

Scan to open on Studocu

Studocu is not sponsored or endorsed by any college or university
Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

(40 MARKS)

SECTION 1: MULTIPLE CHOICE QUESTIONS

1. Key informants, study subjects, focus groups and interviews can be viewed as
   a) Primary data
   b) Secondary data
   c) Data collection tools
   d) Data collection methods
2. Weight, height and length can be described as which measurement of variables?
   a) Discrete variables
   b) Continuous variables
   c) Qualitative variables
   d) Measuring variables
3. Any numerical value that describes a sample is called?
   a) Parameter
   b) Sample
   c) Statistic
   d) Statistics
4. If the sample size is larger, the sampling error is?
   a) Smaller
   b) Larger
   c) Medium
   d) Nil
5. In statistics, continuous data can be classified into the following except?
   a) Ratio scale
   b) Interval scale
   c) Both ratio and interval scale
   d) Ordinal scale
6. What is the geometric mean of these numbers? 8,9,10,11 and 12
   a) 10
   b) 11
   c) 9
   d) 9.8

   $\sqrt[5]{8 \times 9 \times 10 \times 11 \times 12}$

   $\sqrt[5]{95{,}040}$

   $= 9.8$
7. Which of the following statement explains a rating schedule interview?
   a) The type of schedule having questions which guides an observer systematically
   b) Formulated for surveyor to guide him on his information collection
   c) Questions with structured answers to guide on interviews
   d) Questions that helps the investigator to measure the attitude and behavior of an individual
8. Example of stratified sampling method that each group has same sampling fraction is?
   a) Equal sampling
   b) Random sampling
   c) Disproportionate sampling
   d) Proportionate sampling

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

9. If the population is hard to access, which of the sampling method can be used to recruit the respondents?
   a) Random sampling
   b) Snowballing
   c) Judgmental
   d) Quota
10. If the percentage occupancy is more than 100% then,
   a) There are excess bed days
   b) There are vacant bed days
   c) There is underutilization of hospital beds
   d) Percentage occupancy can never be more than 100%
11. A leap year is_____
   a) A year divided by 100
   b) A year divided by 2 and 4
   c) A year ending with 0 and 5
   d) A year that one day is added to in one of its months
12. Inter-ward transfers in the hospital affects_____
   a) Hospital statistics
   b) Ward statistics
   c) Hospital discharge
   d) Transfer out
13. The row headings are also known as _____
   a) Caption
   b) Head note
   c) Row tittle
   d) Stabs
14. Which of the following graphs represents a measure of dispersion?
   a) Bar graph
   b) Scatter graph
   c) Lorenz curve
   d) Z-chart
15. Type of classification made with respect to places is:
   a) Spatial classification
   b) Temporal classification
   c) Geographical classification
   d) Space classification

Geographical
Chronological
Qualitative
Quantitative.

16. The form of grouping where both lower and upper units fall is known as
   a) Exclusive form of grouping
   b) Interval form of grouping
   c) Inclusive form of grouping
   d) Cyclic form of grouping

(10 - 20)
different

17. Which of the following is the best measure of central tendency?
   a) Mean
   b) Median

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

c) Harmonic mean
d) Geometric mean

18. The symbol, μ is an example of _____
a) Population mean
b) Population standard deviation
c) Population variance
d) Sample mean

19. When calculating median, all items of a series are arranged in:
a) Only descending order
b) Only ascending order
c) Descending or ascending order
d) Order magnitude

20. Which methods are used to find mode in a discrete series?
a) Grouping method and ascending method
b) Grouping method and descending method
c) Grouping method and inspecting method
d) Grouping table and inspecting table

21. While calculating the standard deviation, the deviations are only taken from_____
a) Mode
b) Mean
c) Medium
d) Mean deviation

22. Which of the following statement is true?
a) The value of standard deviation can never be Negative
b) The value of standard deviation can never be Positive
c) The value of standard deviation can never be Larger than variance
d) The value of standard deviation can never be Zero

23. The difference between the largest and smallest values in a series is called
a) Quartile deviation
b) Variance
c) Range
d) Inter quartile deviation

24. The 50th percentile is the same as
a) Third quartile
b) First quartile
c) Median
d) Fifth deal

25. Which of the following formula is used to calculate the coefficient of quartile deviation.
a) $\frac{q3-q1}{q3+q1}$
b) $\frac{q3+q1}{4}$
c) $\frac{q3+q1}{q3-q1}$
d) $\frac{q2+q1}{q2-q1}$

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

26. What is the mean deviation for the data given?
20,19,21,26,34,16,17
   a) 18.73
   b) 19.27
   c) 21.86
   d) 21.76

27. The coefficient of variation of ten students in of class is 30 and the variance of 24. What is the arithmetic mean of the students?
   a) 1.63
   b) 16.83
   c) 163
   d) 0.163

28. The scatter within a distribution that is very high on each side indicates_____
   a) High uniformity of data
   b) Outliers of data
   c) Low uniformity of data
   d) Flexibility of data

29. Which of the following statements is true?
   a) If the distribution is negatively skewed, then the Mean is more than the mode
   b) If the distribution is negatively skewed, then the Medium is at right to the mode
   c) If the distribution is negatively skewed, then the Mean is less than the mode
   d) If the distribution is negatively skewed, then the Mean is at right to the median

30. If the frequency distribution has mean +50 and median=48, what is the distribution is.
   a) Positively skewed
   b) Normal
   c) Negatively skewed
   d) Left skewed

31. The relationship between mode, mean and median can be determined by which formula?
   a) Mode= 3mean- 2 median
   b) Mode=3 median – 2mean
   c) Mode= 2 mean-3median
   d) Mean= median=mode

mean − mode = 3(mean − median)
3median = mode + 2mean

32. Which of the following statement is true about Bowley's coefficient of skewness?
   a) It lies between 0 and 1
   b) It lies between -2 and +2
   c) It lies between -1 and 0
   d) It lies between -1 and 1

33. The reciprocal of Arithmetic mean is called_____
   a) Geometric mean
   b) Harmonic mean
   c) Simple average
   d) Weighted mean

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

34. Which of the following formula shows the relationship between Arithmetic mean, Geometric mean and Harmonic mean?
   a) AM> = HM>=GM
   b) GM>= AM>=HM
   c) AM>= GM>=HM
   d) GM>= HM>=AM

35. Which of the following averages is best suited when finding average speed?
   a) Geometric mean
   b) Harmonic mean
   c) Arithmetic mean
   d) Mode

36. What is the HM of the following numbers?
   2,5,7,9
   a) 4.19
   b) 0.953
   c) 4
   d) 5

37. What is the GM of the following numbers?
   8,9,10,11,12
   a) 11
   b) 10
   c) 9
   d) 9.8

38. Which of the following is true about a Geometric
   a) It is the reciprocal of Arithmetic mean
   b) It is the nth root
   c) It is the Square root
   d) It is the Cube root

39. If 7% is added to each value of an observation, the GM of the new variable will be added by
   a) 14%
   b) 3.5%
   c) 7%
   d) 21%

40. It is impossible to find _____ of any value in a data set that is negative
   a) Arithmetic Mean
   b) Harmonic Mean
   c) Geometric Mean
   d) Mode

SECTION 2: SHORT ANSWER QUESTIONS (40MARKS

41. Highlight the purpose of hospital administration statistics (5marks)

42. Suppose there are 50 students in a class where 10 students have average marks of 75% in a subject and 40 students whose average marks in a subject is 60%, calculate the weighted average score for the class. (5marks)

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

$40.5 + \left(\frac{5}{25} \times 10\right) - 40.5 + \left(\frac{4}{25} \times 10\right)$

$42.5 - 42 = 42.25$

43. State five disadvantages of a questionnaire as method of data collection (5marks)
44. Highlight the properties of standard deviation (5marks)
45. In the year 2010, a hospital recorded 20,000 patient's attendances where 45% of them were male. In 2011, there was a noted decrease by 950 patients among the females and an increase of 25% among the males. In 2012, there was an outbreak of cholera which increased hospital attendance by 35% among the females and 25% among the males, from the previous year, present this data in tabulation form observing the principles of table construction (5marks)

46. Compute the Median using the data given below (5marks)

Score	x	No. of students f	cf	fx
10-20	15	10	10	
20-30	25	13	23	
30-40	35	20	43	
40-50	45	25	68	
50-60	55	12	80	
60-70	65	9	89	
70-80	75	8	97	
		97		

$m = L + \left(\frac{\frac{\Sigma f}{2} - cf}{f}\right) c$

$40.5 + \left(\frac{\frac{97}{2} - 43}{68}\right) 10 = 41.3088$

42.7

25

12
14
216

uni, by, Tri, multy

47. Define mode as used in statistics and briefly explain four types of mode (8marks)
48. Highlight the advantages of arithmetic mean (5marks)

35

SECTION 3: LONG ANSWER QUESTIONS (20MARKS)

49. The following information is about the marks obtained by 50 students in an end of semester examination. Calculate the mean using assumed mean method (10marks)

Group	x	Frequency f	$d = x - A$	fd
0-10	5	9	-20	-180
10-20	15	10	-10	-100
20-30	25	18	0	0
30-40	35	6	10	60
40-50	45	7	20	140
		50	30	-80

$\bar{x} = A \pm \frac{\Sigma fd}{\Sigma f}$

$= 25 \pm \frac{-80}{50}$

$25 - 1.6 = 23.4$

14
12
216

50. Compute the following statistical measures using the data below (10marks)

Group	Frequency
0-10	20
10-20	25
20-30	30
30-40	18
40-50	10

a) 40th Percentile
b) 7th Percentile

$median = L_m + \left(\frac{\frac{\Sigma f}{2} - cf_{cb}}{f}\right) c$

$mode = L_m + \left(\frac{D_1^f}{D_1 + D_2}\right) c$

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)